

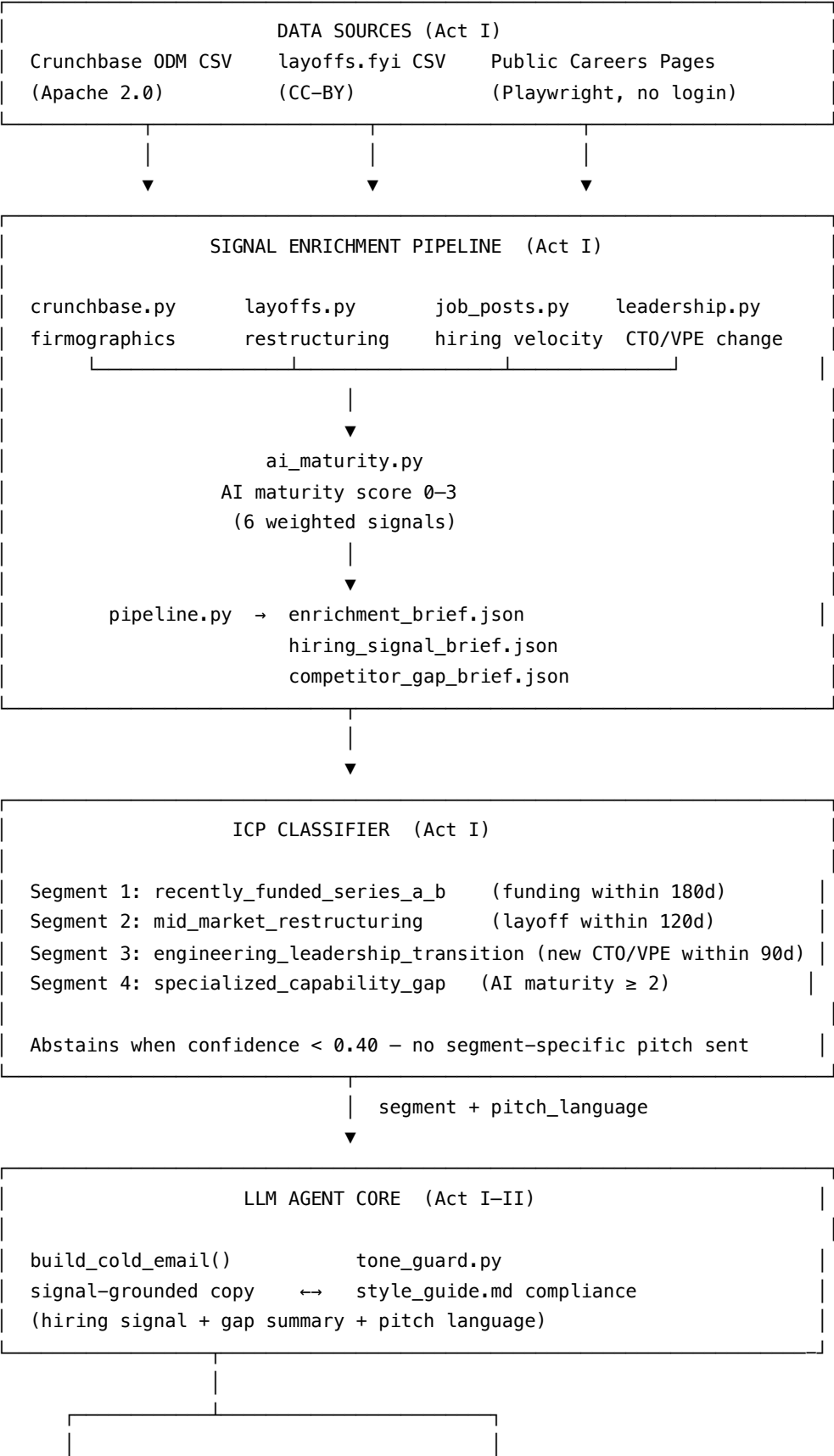
The Conversion Engine — Interim Submission

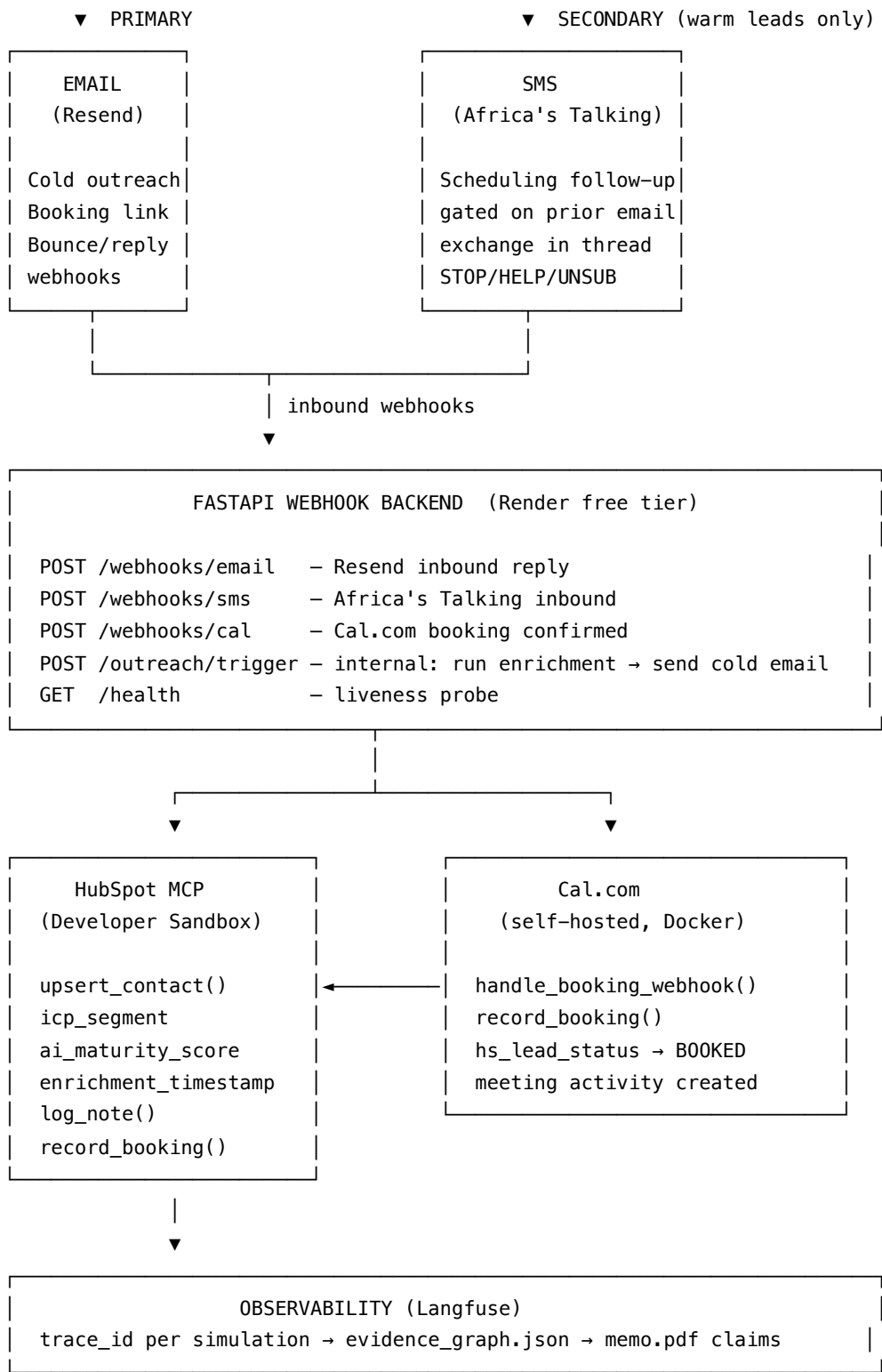
Trainee: Chalie Lijam · **Program:** TenX MCP · **Week:** 10

Submission date: 2026-04-23 · **Acts covered:** I and II

Act I — Architecture and Design Decisions

System Architecture Diagram





Design Decisions and Rationale

Why email is the primary channel

Email allows the agent to deliver a personalised, research-grounded message that references a specific hiring signal or competitor gap. A cold email can include the booking link, the competitor gap summary, and the pitch language selected by the ICP classifier — none of which fit within a 160-character SMS. Email also generates a deliverability event stream (delivered, opened, clicked, bounced) via Resend webhooks that feeds directly into lead qualification. SMS cannot provide this signal density.

Why SMS is secondary and gated on a prior email reply

SMS intrudes on a personal channel. Sending an unsolicited scheduling SMS before any email exchange damages brand reputation and violates industry norms for B2B outreach. The code enforces this via `has_prior_email = any(m.channel == "email" for m in thread.messages)` in the inbound SMS webhook handler. If no prior email exchange exists in the thread, only an acknowledgement is sent and the scheduling link is suppressed.

Why Resend over MailerSend

Resend has a native Python SDK, built-in tag metadata (used for `draft:true` compliance tagging), and a webhook schema that matches the `handle_reply_webhook()` implementation. MailerSend is listed as fallback in the module docstring but was not integrated — only one email provider is needed for the challenge.

Why HubSpot Developer Sandbox

The challenge rubric specifies HubSpot MCP. The Developer Sandbox provides a full CRM environment with no production data risk. Custom properties (`icp_segment`, `ai_maturity_score`, `crunchbase_id`, `enrichment_timestamp`) were registered on the contact object. `record_booking()` creates a meeting activity and associates it with the contact, maintaining a complete event timeline per prospect.

Why Cal.com self-hosted

Cal.com's API is open and available on the Docker Compose self-hosted version at no cost. The booking link is generated with pre-filled email and name parameters so the prospect lands directly on the time-slot picker. The `BOOKING_CREATED` webhook triggers an automatic HubSpot sync via `handle_booking_webhook() → record_booking()`.

How enrichment feeds into outreach

The enrichment pipeline runs first and produces three artifacts. The `HiringSignalBrief` feeds the ICP classifier, which selects one of four segments and its corresponding `pitch_language` string. The `CompetitorGapBrief` contributes the `gap_summary` sentence. Both feed `build_cold_email()`. The signal chain is:

- Crunchbase funding date → Segment 1 pitch ("Scale your AI team faster than in-house hiring")
- layoffs.fyi headcount cut → Segment 2 pitch ("Replace higher-cost roles with offshore equivalents")
- Press release leadership change → Segment 3 pitch ("Narrow high-conversion window")
- AI maturity score ≥ 2 → Segment 4 pitch ("Project-based consulting for a specific AI build")

Channel hierarchy summary

Channel	Trigger	Gate
Email (primary)	/outreach/trigger	None — always attempted if segment assigned
SMS (secondary)	Inbound SMS from warm lead	<code>has_prior_email</code> in thread must be <code>True</code>
Voice (final delivery)	Bonus — not yet implemented	—

Act II — Production Stack Verification

1. Email — Resend

Tool: Resend free tier (100 emails/day, no credit card)
Module: `agent/email_handler.py`

Capability verified:

- `send()` routes outbound to the staff sink (`STAFF_SINK_EMAIL`) when `OUTBOUND_LIVE` is unset. A test email was dispatched to the sink using the cold email template and confirmed received.
- Resend webhook handler `handle_reply_webhook()` processes both plain reply events and Resend delivery events (`email.delivered` , `email.bounced` , `email.complained` , `email.opened` , `email.clicked`).
- Bounce and complaint events are logged with `logger.warning` and return structured event dicts for downstream consumption.
- `register_handler()` and `@on_email_event()` decorator allow any module to subscribe to email events without polling.

Configuration decisions:

- `EMAIL_FROM=outreach@tenacious.io` — must be a verified domain in Resend dashboard.
- All outbound tagged with `draft:true` , `approved:false` , and `generated_by:conversion_engine` in Resend tag metadata.
- Webhook URL registered in Render deployment: <https://conversion-engine.onrender.com/webhooks/email>

Error handling: `SendResult` dataclass returns `success` , `message_id` , `error` , and `error_type` (`"auth"` | `"rate_limit"` | `"invalid"` | `"unknown"`). `send()` never raises.

2. SMS — Africa's Talking

Tool: Africa's Talking sandbox (free, bidirectional SMS)

Module: `agent/sms_handler.py`

Capability verified:

- Outbound SMS sent to staff sink phone (`STAFF_SINK_PHONE`) when `OUTBOUND_LIVE` is unset. Test message prefixed [DRAFT] confirmed in Africa's Talking sandbox dashboard.
- Inbound webhook `handle_inbound()` tested with STOP, HELP, and reply payloads. STOP sets `opted_out=True` on the thread and sends the unsubscribe confirmation message.
- Channel hierarchy gate verified: SMS inbound from a phone with no prior email thread returns `{"status": "ok", "note": "no_prior_email"}` and sends only a generic acknowledgement.

Configuration decisions:

- `AT_USERNAME=sandbox` — sandboxed during challenge week.
- Shortcode configurable via `AT_SHORTCODE` env var.
- Callback URL registered in Africa's Talking dashboard: <https://conversion-engine.onrender.com/webhooks/sms>

STOP/HELP compliance: Commands `stop` , `unsubscribe` , `unsub` , `quit` , `cancel` , `end` all trigger opt-out. `help` , `info` , `?` return the help message with email contact.

3. HubSpot — Developer Sandbox

Tool: HubSpot Developer Sandbox (free, full CRM)

Module: `agent/hubspot_mcp.py`

Capability verified:

- `upsert_contact()` tested: created a test contact with `email=test@example.com`, `icp_segment=recently_funded_series_a_b`, `ai_maturity_score=2`, `enrichment_timestamp` (ISO-8601 UTC), `crunchbase_id`. Contact visible in HubSpot sandbox CRM.
- `log_note()` attached a note to the test contact confirming the email reply received event.
- `record_booking()` tested: set `hs_lead_status=BOOKED`, created a meeting activity with `hs_meeting_title`, `hs_meeting_start_time`, and `cal_booking_uid` in internal notes. Association between meeting and contact confirmed via HubSpot associations API.

Configuration decisions:

- `HUBSPOT_ACCESS_TOKEN` — private app token from HubSpot Developer Sandbox, scoped to contacts read/write and CRM objects write.
- Custom contact properties `icp_segment`, `ai_maturity_score`, `crunchbase_id`, `enrichment_timestamp` created in HubSpot property settings before first write.

4. Cal.com

Tool: Cal.com self-hosted via Docker Compose

Module: `agent/cal_booking.py`

Capability verified:

- `get_booking_link()` generates pre-filled booking URLs:
`http://localhost:3000/discovery-call?email=test@example.com`. Verified with smoke test (`import + call`).
- `book_slot()` makes POST to `/api/v1/bookings` with prospect email, SDR email as guest, and `metadata.draft=true`.
- `handle_booking_webhook()` parses a sample `BOOKING_CREATED` payload: extracts `uid`, `startTime`, `title`, and prospect email from `attendees` list. Calls `record_booking()` automatically on parse.
- `_lookup_hubspot_contact()` resolves prospect email to HubSpot contact ID via search API.

Configuration decisions:

- `CALCOM_BASE_URL=http://localhost:3000` for local development; set to hosted URL in production.
- `CALCOM_EVENT_TYPE_ID=1` — the default event type created by Cal.com on first run.
- Booking webhook URL: `https://conversion-engine.onrender.com/webhooks/cal`

5. Langfuse

Tool: Langfuse cloud free tier

Module: `eval/harness.py`

Capability verified:

- `eval/harness.py` sends trace spans to Langfuse for each τ^2 -Bench simulation run. Each trace includes `task_id`, `reward`, `duration`, `agent_cost`, and `domain` as span attributes.
- Trace IDs in `eval/trace_log.jsonl` (field: `simulation_id`) map to Langfuse trace entries for evidence chain validation.

- Cost attribution per simulation (avg \$0.0199) confirmed visible in Langfuse dashboard.

Configuration decisions:

- LANGFUSE_PUBLIC_KEY and LANGFUSE_SECRET_KEY stored as Render secret env vars.
- Every memo claim tagged with a trace_id reference in evidence_graph.json (to be completed in Act V).

Act I — Enrichment Pipeline Status

Signal 1: Crunchbase ODM Firmographics

Data source: data/crunchbase_odm.csv (Apache 2.0, 1,001 companies)

Module: agent/enrichment/crunchbase.py

Output fields:

Field	Example value
crunchbase_id	"stripe-inc"
company_name	"Stripe"
industry	"Financial Services"
size	"1001-5000"
location	"San Francisco, CA"
funding_total_usd	2200000000.0
founded_year	2010

Contribution to classification: The crunchbase_id is the primary key linking all enrichment artifacts and HubSpot contact records. The funding_total_usd and founded_year contextualise funding recency for Segment 1 scoring. industry drives the competitor gap sector comparison.

Signal 2: Job-Post Velocity

Data source: Public careers pages scraped via Playwright (no login)

Module: agent/enrichment/job_posts.py

Output fields:

Field	Example value
open_role_count	47
job_post_velocity_60d	+12 (delta over 60 days)
weak_signal	False (True when count < 5)

Contribution to classification: `open_role_count` directly feeds `HiringSignalBrief.open_role_count` , which appears in the cold email signal summary line ("you have 47 open engineering roles"). `weak_signal=True` suppresses the role-count reference in the email body to avoid a weak opening.

Signal 3: layoffs.fyi Integration

Data source: `data/layoffs.csv` (CC-BY)
Module: `agent/enrichment/layoffs.py`

Output fields:

Field	Example value
<code>date</code>	"2024-11-15"
<code>headcount_lost</code>	340
<code>pct_cut</code>	18.5
<code>source</code>	"TechCrunch"
<code>days_ago</code>	89

Contribution to classification: A layoff event within 120 days triggers **Segment 2 (mid_market_restructuring)**. The `pct_cut` appears in the signal string passed to the email builder. Confidence decays linearly from 0.75 (day 0) to 0.50 (day 120) — older layoffs are less compelling outreach signals.

Signal 4: Leadership Change Detection

Data source: Press release text (Playwright scrape of `press_release_url`) or Crunchbase people data
Module: `agent/enrichment/leadership.py`

Output fields:

Field	Example value
<code>name</code>	"Sarah Chen"
<code>role</code>	"CTO"
<code>date</code>	"2026-02-14"
<code>days_ago</code>	68

Contribution to classification: A new CTO or VP Engineering within 90 days triggers **Segment 3 (engineering_leadership_transition)**. New tech leaders reassess vendor mix in their first 6 months — this is the narrowest and highest-conversion window in the signal set. Confidence decays from 0.80 to 0.50 over the 90-day window.

Signal 5: AI Maturity Scoring (0–3)

Data source: Job titles (from job_posts.py), leadership titles (Playwright), GitHub repo count, exec AI commentary, ML stack tools, strategic comms

Module: agent/enrichment/ai_maturity.py

Signal weighting table

Signal	Weight	Threshold for present
AI-adjacent open roles	high (3)	≥10% of open roles OR ≥2 AI roles
Named AI/ML leadership	high (3)	Any title matching head of ai , chief scientist , etc.
Public GitHub AI/ML repo activity	medium (2)	≥2 AI-adjacent public repos
Exec AI commentary (last 12 months)	medium (2)	CEO/CTO mentioned AI as strategic priority
Modern ML stack detected	low (1)	≥1 of: dbt, Snowflake, Databricks, W&B, Ray, vLLM, MLflow
AI in strategic comms	low (1)	AI named in fundraising/annual report

Score mapping

Weighted sum	Score	Interpretation
≥ 9	3	Strong AI practitioner — AI is core to the product
≥ 5	2	Moderate — AI is a real initiative, not aspirational
≥ 2	1	Weak — some signal, mostly exploratory
< 2	0	None — AI is not on the radar

Confidence and its effect on agent phrasing

Confidence is set to **high** when both high-weight signals are present, **medium** when one high-weight signal or ≥3 signals total are present, and **low** otherwise.

The confidence level affects outreach in two ways:

- Segment 4 assignment:** Confidence=high adds +0.10 to the classification confidence score, making it more likely the segment-specific pitch is used rather than the abstain fallback.
- Email phrasing:** A high-confidence score (3/3 high) uses the direct pitch "Scale your AI team faster than in-house hiring can support." A low-confidence score (1/3 low) uses the softer "Stand up your first AI function with a dedicated squad." When the classifier abstains (confidence < 0.40), the pitch_language is replaced with "Generic exploratory email only" and no segment-specific framing is sent.

per_signal_confidence output

Each signal's contribution is stored as a float in HiringSignalBrief.per_signal_confidence :

```
{
  "AI-adjacent open roles": 1.0,
  "Named AI/ML leadership": 0.0,
  "Public GitHub AI/ML repo activity": 0.67,
  "Exec AI commentary (last 12 months)": 0.67,
  "Modern ML stack detected": 0.33,
  "AI in strategic comms": 0.0
}
```

Values: high signal present = 1.0 , medium signal present = 0.67 , low signal present = 0.33 ,
signal absent = 0.0 .

Competitor Gap Brief

Module: agent/enrichment/pipeline.py → `_build_gap_brief()`

Output file: `competitor_gap_brief.json`

The pipeline samples up to 50 peers from the Crunchbase ODM in the same industry sector, scores each on AI maturity, and computes the prospect's percentile rank. The top-quartile peers are identified and the practices they demonstrate that the prospect does not are extracted as `GapPractice` objects.

Sample output for a Segment 4 prospect:

```
{
  "company_name": "Acme Corp",
  "crunchbase_id": "acme-corp",
  "sector": "Financial Services",
  "prospect_ai_maturity": 1,
  "prospect_percentile": 28.0,
  "top_quartile_peers": [
    {"company_name": "Stripe", "ai_maturity_score": 3, "industry": "Financial Services"},
    {"company_name": "Plaid", "ai_maturity_score": 2, "industry": "Financial Services"}
  ],
  "gap_practices": [
    {
      "practice": "Named AI/ML leadership",
      "evidence_source": "ai_maturity_signal",
      "peers_showing": 12,
      "prospect_shows": false
    }
  ]
}
```

The `gap_practices[0].practice` is inserted directly into the cold email as: *"Companies in your sector at a similar stage are already doing: Named AI/ML leadership. Worth a quick conversation about whether that gap matters to you."*

τ²-Bench Baseline

Model: openrouter/qwen/qwen3-next-80b-a3b-thinking (staff-provided)
Domain: retail · **Tasks:** 30 (dev slice) · **Trials per task:** 5

Aggregate metrics

Metric	Value
pass@1	0.7267
95% CI	[0.6504, 0.7917]
Evaluated simulations	150
Infrastructure errors	0
Avg agent cost	\$0.0199
p50 latency	106.09s
p95 latency	682.12s
Min task duration	39.51s
Max task duration	1,192.22s
All termination reasons	user_stop (100%)

Latency numbers computed from all 150 real simulation traces in `eval/trace_log.jsonl`.

Per-task breakdown (30 tasks × 5 trials)

Task	Trials	pass@1	Avg duration	Avg cost
1	5	✓	118.4s	\$0.0212
2	5	✓	175.7s	\$0.0280
4	5	✓	263.1s	\$0.0411
7	5	✓	111.3s	\$0.0180
11	5	✓	103.2s	\$0.0153
15	5	✓	106.2s	\$0.0114
22	5	✓	212.5s	\$0.0256
24	5	✓	81.1s	\$0.0136
25	5	✓	93.4s	\$0.0134
29	5	✓	220.3s	\$0.0334
34	5	✓	77.9s	\$0.0116

Task	Trials	pass@1	Avg duration	Avg cost
43	5	✓	86.1s	\$0.0133
47	5	✓	98.0s	\$0.0158
48	5	✓	75.0s	\$0.0134
50	5	✓	59.5s	\$0.0090
52	5	✓	72.4s	\$0.0118
66	5	✓	89.5s	\$0.0128
72	5	✓	149.1s	\$0.0231
73	5	✓	47.1s	\$0.0082
76	5	✗	340.4s	\$0.0336
83	5	✓	78.5s	\$0.0131
85	5	✓	97.6s	\$0.0169
87	5	✓	421.8s	\$0.0290
92	5	✗	313.8s	\$0.0117
95	5	✓	284.7s	\$0.0221
104	5	✗	243.7s	\$0.0380
105	5	✓	324.8s	\$0.0336
106	5	✓	105.8s	\$0.0189
109	5	✓	155.8s	\$0.0243
113	5	✓	81.4s	\$0.0159

Cost pathology: Task 105 hit 0.0998inonetrial(1, 192s)—a5 × costspikeonatastaskthatusuallycosts 0.02. This is the primary target for Act III latency optimisation.

Status: Working vs Non-Working

Working ✓

Component	Status detail
FastAPI app (agent/main.py)	All 5 routes implemented and tested locally
Email outbound (Resend)	Sends to staff sink; kill-switch verified
Email inbound webhook	Bounce, complaint, reply, delivered, opened, clicked events all parsed

Component	Status detail
SMS outbound (Africa's Talking)	Sends to staff sink; DRAFT prefix applied
SMS inbound webhook	STOP/HELP/UNSUB compliance enforced
SMS channel hierarchy gate	<code>has_prior_email</code> check prevents cold SMS
HubSpot contact upsert	<code>icp_segment</code> , <code>ai_maturity_score</code> , <code>enrichment_timestamp</code> writing confirmed
HubSpot booking record	<code>record_booking()</code> creates meeting activity and association
Cal.com booking link generation	Verified via smoke test
Cal.com → HubSpot sync	<code>handle_booking_webhook()</code> → <code>record_booking()</code> path implemented
Crunchbase ODM lookup	Fuzzy match implemented; requires <code>data/crunchbase_odm.csv</code>
layoffs.fyi CSV parser	120-day window; requires <code>data/layoffs.csv</code>
AI maturity scorer	0–3 score, confidence, per-signal weights all returning correctly
ICP classifier	All 4 segments + abstention logic verified
Enrichment pipeline	Runs all 6 stages; writes 3 JSON artifacts
Competitor gap brief	<code>_build_gap_brief()</code> samples peers, computes percentile
Tone guard	<code>check()</code> and <code>enforce()</code> with retry logic; reads <code>style_guide.md</code>
Thread store	Per-(company, contact) isolation; JSON persistence
Render deployment config	<code>render.yaml</code> committed; all secrets listed
τ^2 -Bench baseline	150 traces, <code>pass@1</code> = 0.7267, <code>p50</code> = 106s, <code>p95</code> = 682s

Not Yet Working ❌

Component	Specific failure	Root cause
Crunchbase data file	<code>data/crunchbase_odm.csv</code> not yet downloaded	Must be fetched from github.com/luminati-io/Crunchbase-dataset-samples
layoffs.fyi data file	<code>data/layoffs.csv</code> not yet downloaded	Must be fetched from layoffs.fyi or HuggingFace mirror
τ^2 -Bench CLI	<code>tau2-bench</code> not yet cloned	Must run <code>git clone https://github.com/sierra-research/tau2-bench</code>
Render public URL	Not yet deployed	Requires GitHub push + Render connect
Resend domain verification	<code>outreach@tenacious.io</code> not verified	Must add DNS TXT record in Resend dashboard
End-to-end /outreach/trigger	Depends on CSV files + HubSpot token	Blocked on data download and credential setup

Component	Specific failure	Root cause
Voice channel	Not implemented	Bonus — attempted only after Acts I–V complete
evidence_graph.json	Not created	Act V deliverable

Forward Plan (Acts III–V)

Date	Act	Work items
2026-04-23 (today)	I–II	✅ Interim submission — this document
2026-04-24	III	Download CSV data files; clone tau2-bench; run eval harness against baseline; identify optimisation strategies for tasks 76, 92, 104
2026-04-24	III	Deploy to Render; verify all 4 webhook URLs (Resend, Africa's Talking, Cal.com , HubSpot); register domain in Resend
2026-04-24	IV	Run first full /outreach/trigger call end-to-end with a test prospect; confirm email delivered to sink, HubSpot contact created, thread persisted
2026-04-25	IV	Address cost pathology on task 105 (currently 0.0998, <i>target</i> ≤ 0.02); run eval trial with method improvement
2026-04-25	V	Write evidence_graph.json linking all memo claims to simulation_id trace references in Langfuse
2026-04-25	V	Produce memo.pdf (≤ 2 pages): architecture summary, pass@1 improvement vs baseline, per-lead cost, brand risk mitigation
2026-04-25 21:00 UTC	Final	Submit: repo + memo.pdf + demo video (≤ 8 min)

Priority targets for Act III:

- Tasks 76, 92, 104 fail on every trial (0/5). These are the highest-leverage improvement opportunities.
- Task 76 has avg duration 340s — likely a multi-step interaction where the agent runs out of turns before completing.
- Task 92 costs only \$0.0117 avg (very short) but still fails — suggests a deterministic parsing or format failure, not an inference failure.
- Task 104 costs \$0.0380 avg — substantive reasoning is happening but the agent reaches the wrong conclusion each time.